

Reinforcement Learning

Chapter 2: Model-based RL

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Bellman Equation: Backup Diagram

Bellman equation gives an

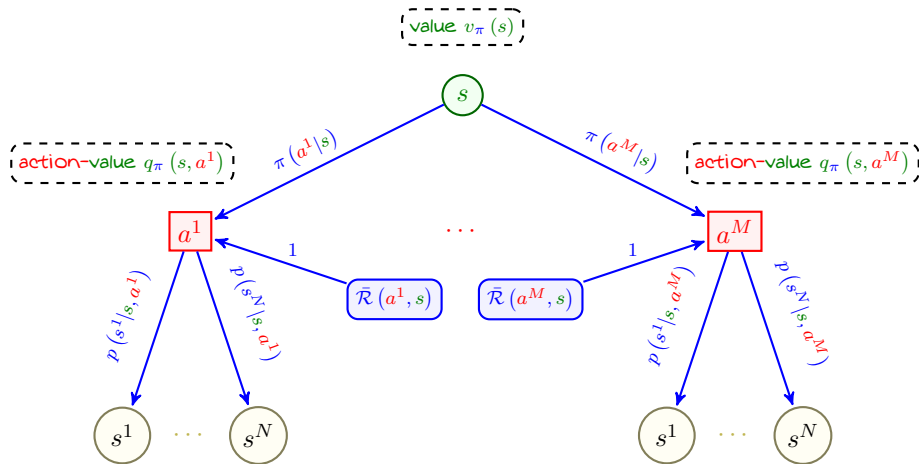
interesting *visualization* for *values* and *action-values*

which can be shown in the so-called *backup diagram*

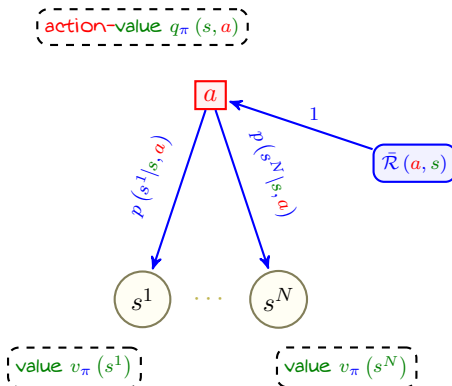
For simplicity, we consider $\gamma = 1$ in the *backup diagram*

- Each circle node is a *state* and carries the *value of the state*
- Each square node is an *action* and carries the *action-value of the pair*
- Each *edge* is a transition and carries *a probability*
- As we pass from *leaves* to *root*
 - Value of each node multiplies to its probability on the edge
 - They add up when they meet at a parent node

Backup Diagram: For Given Policy



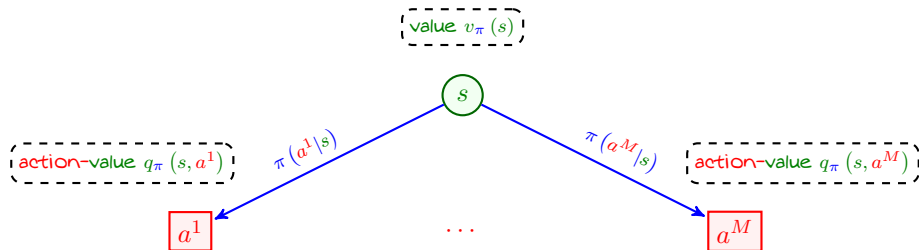
Backup Diagram: For Given Policy



Let's look at it part by part: *first we pass from leaves to **action** parent*

$$q_\pi(s, a) = \bar{\mathcal{R}}(s, a) + \sum_{n=1}^N v_\pi(s^n) p(s^n|s, a)$$

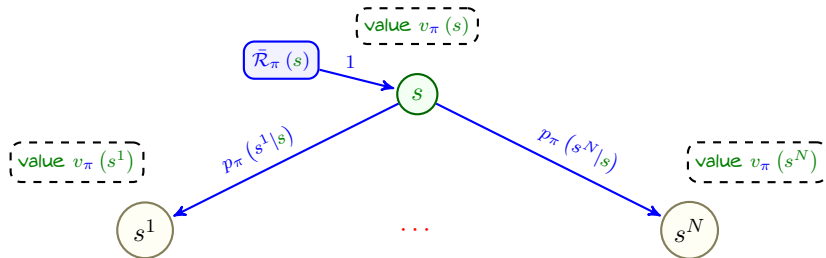
Backup Diagram: For Given Policy



Then, we pass from **action** parents to the **root state**

$$v_{\pi}(s) = \sum_{m=1}^M \pi(a^m|s) q_{\pi}(s, a^m)$$

Backup Diagram: For Given Policy



We could also have its alternative form **expected** over **actions**

$$v_{\pi}(s) = \bar{\mathcal{R}}_{\pi}(s) + \sum_{n=1}^N p_{\pi}(s^n|s) v_{\pi}(s^n)$$

Optimal Value: Optimality Constraint

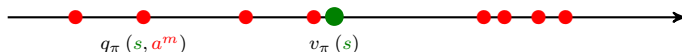
In Assignment 1, you show that for any *state* we have

$$v_{\pi}(s) = \sum_{m=1}^M q_{\pi}(s, a^m) \pi(a^m | s)$$

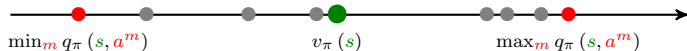
Now, recall that *policy* is a conditional *distribution* meaning that

$$0 \leq \pi(a^m | s) \leq 1$$

We can think of it as



Optimal Value: Optimality Constraint



It is hence obvious that

$$\min_m q_\pi(s, a^m) \leq v_\pi(s) \leq \max_m q_\pi(s, a^m)$$

*We can use this simple fact to find a constraint on **optimal values***

Optimal Value: Optimality Constraint

If our policy is the *optimal policy*; then, we should have

$$v_{\star}(s) = \text{maximum possible value} = \max_m q_{\star}(s, a^m)$$

- + But, can we guarantee that we can achieve such value?
- Sure! We can set an *optimal policy* to

$$\pi^{\star}(a^m | s) = \begin{cases} 1 & m = \operatorname{argmax}_m q_{\star}(s, a^m) \\ 0 & m \neq \operatorname{argmax}_m q_{\star}(s, a^m) \end{cases}$$

- + They both depend on $q_{\star}(s, a^m)$! We don't have the *optimal action-values*!
- Sure! But, we could say that *optimal values must* satisfy this constraint: if not, they cannot be *optimal*

Optimal Value: Optimality Constraint

Optimality Constraint

Optimal value at each state s satisfies the following identity

$$v_{\star}(s) = \max_m q_{\star}(s, a^m)$$

and is achieved if we set the policy to

$$\pi^{\star}(a^m | s) = \begin{cases} 1 & m = \operatorname{argmax}_m q_{\star}(s, a^m) \\ 0 & m \neq \operatorname{argmax}_m q_{\star}(s, a^m) \end{cases}$$

which is an *optimal* policy

- + But, how can we relate this constraint to *Bellman equation*?
- Let's see!

Optimal Value: *Bellman Equation*

We know from Bellman equation II for **action-value** function that

$$q_{\pi}(s, a) = \bar{\mathcal{R}}(s, a) + \gamma \sum_{n=1}^N v_{\pi}(s^n) p(s^n | s, a)$$

If we play with **optimal policy**: *we are going to have same identity*

$$q_{\star}(s, a) = \bar{\mathcal{R}}(s, a) + \gamma \sum_{n=1}^N v_{\star}(s^n) p(s^n | s, a)$$

We now substitute it in **optimality constraint**

$$v_{\star}(s) = \max_m \left[\bar{\mathcal{R}}(s, a^m) + \gamma \sum_{n=1}^N v_{\star}(s^n) p(s^n | s, a^m) \right]$$

Optimal Value: Bellman Equation

This is again a *recursive equation* that

does **not** depend on any policy!

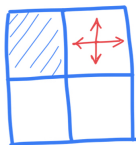
Bellman Optimality Equation

The optimal value function $v_{\star}(s)$ satisfies

$$v_{\star}(s) = \max_m \left[\bar{\mathcal{R}}(s, a^m) + \gamma \sum_{n=1}^N v_{\star}(s^n) p(s^n | s, a^m) \right]$$

We can again treat it as a fixed-point equation and solve it for $v_{\star}(s)$

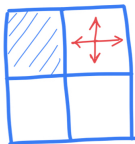
Example: Dummy Grid World



Let's find optimal values for our dummy grid world: we first find $\bar{\mathcal{R}}(s, a)$

$$\begin{array}{llll}
 \bar{\mathcal{R}}(0, a) = 0 & \bar{\mathcal{R}}(1, 0) = -1 & \bar{\mathcal{R}}(2, 0) = -0.5 & \bar{\mathcal{R}}(3, 0) = -1 \\
 \bar{\mathcal{R}}(1, 1) = -1 & \bar{\mathcal{R}}(2, 1) = -0.5 & \bar{\mathcal{R}}(3, 1) = -0.5 & \\
 \bar{\mathcal{R}}(1, 2) = -0.5 & \bar{\mathcal{R}}(2, 2) = -1 & \bar{\mathcal{R}}(3, 2) = -0.5 & \\
 \bar{\mathcal{R}}(1, 3) = -0.5 & \bar{\mathcal{R}}(2, 3) = -1 & \bar{\mathcal{R}}(3, 3) = -1 &
 \end{array}$$

Example: Dummy Grid World

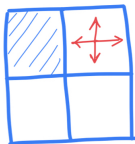


We next write down Bellman equations

- ① Since $s = 0$ is a **terminal state** we know that $v_{\star}(0) = 0$
- ② Now, let's consider $s = 1$

$$\begin{aligned}
 p(0|1, 0) &= 1 \\
 p(1|1, 0) &= 0 \\
 p(2|1, 0) &= 0 \\
 p(3|1, 0) &= 0
 \end{aligned}
 \rightsquigarrow \sum_{\bar{s}=0}^4 v_{\star}(\bar{s}) p(\bar{s}|1, 0) = v_{\star}(0) = 0$$

Example: Dummy Grid World

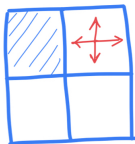


We next write down Bellman equations

- ① Since $s = 0$ is a *terminal state* we know that $v_{\star}(0) = 0$
- ② Now, let's consider $s = 1$

$$\begin{aligned}
 p(0|1, \textcolor{red}{1}) &= 0 \\
 p(1|1, \textcolor{red}{1}) &= 0 \\
 p(2|1, \textcolor{red}{1}) &= 0 \\
 p(3|1, \textcolor{red}{1}) &= 1
 \end{aligned}
 \rightsquigarrow \sum_{\bar{s}=0}^4 v_{\star}(\bar{s}) p(\bar{s}|\textcolor{green}{1}, \textcolor{red}{1}) = v_{\star}(\textcolor{blue}{3})$$

Example: Dummy Grid World

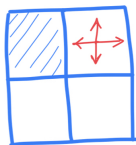


We next write down Bellman equations

- ① Since $s = 0$ is a **terminal state** we know that $v_{\star}(0) = 0$
- ② Now, let's consider $s = 1$

$$\begin{aligned}
 p(0|1, 2) &= 0 \\
 p(1|1, 2) &= 1 \\
 p(2|1, 2) &= 0 \\
 p(3|1, 2) &= 0
 \end{aligned}
 \rightsquigarrow \sum_{\bar{s}=0}^4 v_{\star}(\bar{s}) p(\bar{s}|1, 2) = v_{\star}(1)$$

Example: Dummy Grid World



We next write down Bellman equations

- ① Since $s = 0$ is a *terminal state* we know that $v_{\star}(0) = 0$
- ② Now, let's consider $s = 1$

$$\begin{aligned}
 p(0|1, \textcolor{red}{3}) &= 0 \\
 p(1|1, \textcolor{red}{3}) &= 1 \\
 p(2|1, \textcolor{red}{3}) &= 0 \\
 p(3|1, \textcolor{red}{3}) &= 0
 \end{aligned}
 \rightsquigarrow \sum_{\bar{s}=0}^4 v_{\star}(\bar{s}) p(\bar{s}|\textcolor{green}{1}, \textcolor{red}{3}) = v_{\star}(\textcolor{blue}{1})$$

Example: Dummy Grid World



We next write down Bellman equations

- ① Since $s = 0$ is a **terminal state** we know that $v_{\star}(0) = 0$
- ② Now, let's consider $s = 1$

$$\begin{aligned}
 v_{\star}(1) &= \max_m \left[\bar{\mathcal{R}}(1, a^m) + \sum_{\bar{s}=0}^4 v_{\star}(\bar{s}) p(\bar{s}|1, a^m) \right] \\
 &= \max \{-1, -1 + v_{\star}(3), -0.5 + v_{\star}(1), -0.5 + v_{\star}(1)\}
 \end{aligned}$$

Example: Dummy Grid World



We next write down Bellman equations

- 1 Since $s = 0$ is a **terminal state** we know that $v_{\star}(0) = 0$
- 2 Now, let's consider $s = 1$

$$v_{\star}(1) = \max \{-1, -1 + v_{\star}(3), -0.5 + v_{\star}(1), -0.5 + v_{\star}(1)\}$$

- 3 Similarly, we have for $s = 2$

$$v_{\star}(2) = \max \{-0.5 + v_{\star}(2), -0.5 + v_{\star}(2), -1 + v_{\star}(3), -1\}$$

Example: *Dummy Grid World*



We next write down Bellman equations

- ① Since $s = 0$ is a *terminal state* we know that $v_{\star}(0) = 0$
- ② Now, let's consider $s = 1$

$$v_{\star}(1) = \max \{-1, -1 + v_{\star}(3), -0.5 + v_{\star}(1), -0.5 + v_{\star}(1)\}$$

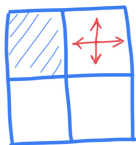
- ③ Similarly, we have for $s = 2$

$$v_{\star}(2) = \max \{-0.5 + v_{\star}(2), -0.5 + v_{\star}(2), -1 + v_{\star}(3), -1\}$$

- ④ Finally for $s = 3$, we have

$$v_{\star}(3) = \max \{-1 + v_{\star}(2), -0.5 + v_{\star}(3), -0.5 + v_{\star}(3), -1 + v_{\star}(1)\}$$

Example: *Dummy Grid World*



After sorting out the Bellman equations, we get

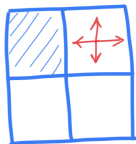
$$v_{\star}(1) = \max \{-1, -1 + v_{\star}(3), -0.5 + v_{\star}(1)\}$$

$$v_{\star}(2) = \max \{-1, -1 + v_{\star}(3), -0.5 + v_{\star}(2)\}$$

$$v_{\star}(3) = \max \{-1 + v_{\star}(2), -0.5 + v_{\star}(3), -1 + v_{\star}(1)\}$$

We should now solve this system of equations

Example: *Dummy Grid World*



We first note that

$$\max \{-1, -1 + v_{\star}(3), -0.5 + v_{\star}(1)\} \neq -0.5 + v_{\star}(1)$$

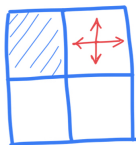
Proof: Assume that

$$\max \{-1, -1 + v_{\star}(3), -0.5 + v_{\star}(1)\} = -0.5 + v_{\star}(1)$$

Then, we have

$$v_{\star}(1) = -0.5 + v_{\star}(1) \rightsquigarrow 0 = -0.5 \quad \text{impossible!}$$

Example: *Dummy Grid World*



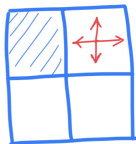
For the same reason, we have

$$\begin{aligned}\max \{-1, -1 + v_{\star}(3), -0.5 + v_{\star}(2)\} &\neq -0.5 + v_{\star}(2) \\ \max \{-1 + v_{\star}(2), -0.5 + v_{\star}(3), -1 + v_{\star}(1)\} &\neq -0.5 + v_{\star}(3)\end{aligned}$$

So the equations reduce to

$$\begin{aligned}v_{\star}(1) &= \max \{-1, -1 + v_{\star}(3)\} = v_{\star}(2) \\ v_{\star}(2) &= \max \{-1, -1 + v_{\star}(3)\} = v_{\star}(1) \\ v_{\star}(3) &= \max \{-1 + v_{\star}(2), -1 + v_{\star}(1)\} = -1 + v_{\star}(1)\end{aligned}$$

Example: *Dummy Grid World*



Thus, we should only solve

$$v_{\star}(1) = \max \{-1, -1 + v_{\star}(3)\}$$

$$v_{\star}(3) = -1 + v_{\star}(1)$$

It is *again easy to see* that $\max \{-1, -1 + v_{\star}(3)\} \neq -1 + v_{\star}(3)$; therefore,

$$v_{\star}(1) = v_{\star}(2) = -1 \rightsquigarrow v_{\star}(3) = -2$$

Well! This is what we expected!

From Optimal Values to Optimal Policy

- + But we are looking for *optimal policy*!
- We can actually back-track *optimal policy*, once we have *optimal value*

The idea is quite simple:

- ① We can find optimal values from *Bellman optimality equations*
- ② We could then find the *optimal action*-values
- ③ We finally get the *optimal policy* from *optimal action*-values

Finding Optimal Policy: *Back-Tracking from Optimal Values*

We could summarize this approach algorithmically as follows

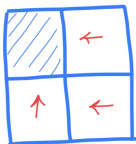
OptimBackTrack():

- 1: **for** $n = 1 : N$ **do**
- 2: Solve Bellman equation $v_\star(s^n) = \max_m [\bar{\mathcal{R}}(s^n, a^m) + \gamma \mathbb{E} \{v_\star(\bar{S}) | s^n, a^m\}]$
- 3: **end for**
- 4: **for** $n = 1 : N$ **do**
- 5: **for** $m = 1 : M$ **do**
- 6: Compute action-value $q_\star(s^n, a^m) = \bar{\mathcal{R}}(s^n, a^m) + \gamma \mathbb{E} \{v_\star(\bar{S}) | s^n, a^m\}$
- 7: **end for**
- 8: Compute optimal policy via optimality constraint

$$\pi^\star(a^m | s) = \begin{cases} 1 & m = \underset{m}{\operatorname{argmax}} q_\star(s, a^m) \\ 0 & m \neq \underset{m}{\operatorname{argmax}} q_\star(s, a^m) \end{cases}$$

9: **end for**

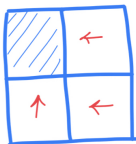
Example: Dummy Grid World



Let's find optimal policy at **state** $s = 1$ in our dummy grid world: first, we write

$$\begin{bmatrix} q_{\star}(1,0) \\ q_{\star}(1,1) \\ q_{\star}(1,2) \\ q_{\star}(1,3) \end{bmatrix} = \begin{bmatrix} \bar{\mathcal{R}}(1,0) + \sum_{\bar{s}} v_{\star}(\bar{s}) p(\bar{s}|1,0) \\ \bar{\mathcal{R}}(1,1) + \sum_{\bar{s}} v_{\star}(\bar{s}) p(\bar{s}|1,1) \\ \bar{\mathcal{R}}(1,2) + \sum_{\bar{s}} v_{\star}(\bar{s}) p(\bar{s}|1,2) \\ \bar{\mathcal{R}}(1,3) + \sum_{\bar{s}} v_{\star}(\bar{s}) p(\bar{s}|1,3) \end{bmatrix} = \begin{bmatrix} -1+0 \\ -1-2 \\ -0.5-1 \\ -0.5-1 \end{bmatrix} = \begin{bmatrix} -1 \\ -3 \\ -1.5 \\ -1.5 \end{bmatrix}$$

Example: *Dummy Grid World*



The optimal policy at **state** $s = 1$ is then given by

$$\pi^*(a|1) = \begin{cases} 1 & a = \underset{a}{\operatorname{argmax}} q_\star(1, a) \\ 0 & a \neq \underset{a}{\operatorname{argmax}} q_\star(1, a) \end{cases} = \begin{cases} 1 & a = 0 \\ 0 & a \neq 0 \end{cases}$$

Well! We know that this is *optimal* in this problem!

Finding Optimal Policy: *Back-Tracking from Optimal Values*

- + Wait a moment! Does that mean that our optimal policy is always *deterministic*? But, you said it could be also *random*!
- Well! In some cases we could find *random optimal policies* as well!

If $q_{\star}(s, a^m)$ has a single maximizer; then,

optimal policy $\pi^*(a^m|s)$ is *deterministic*

But, if it has *multiple* maximizers

optimal policy $\pi^*(a^m|s)$ can also be *random*

Finding Optimal Policy: *General Form*

Generic Optimal Policy

Assume that $m^1, \dots, m^J$ are all maximizers of $q_\star(s, a^m)$; then, policy

$$\pi^\star(a^m|s) = \begin{cases} p_1 & m = m^1 \\ \vdots & \\ p_J & m = m^J \\ 0 & m \notin \{m^1, \dots, m^J\} \end{cases}$$

for any $p_1, \dots, p_J$ that satisfy

$$\sum_{j=1}^J p_j = 1$$

is an *optimal* policy

Finding Optimal Policy

- + But, still we could have a *deterministic optimal* policy in such cases! Right?!
- Sure! We could *always* have a *deterministic optimal* policy!

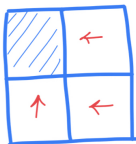
Deterministic Optimal Policy

With known MDP for the environment, there exists *at least one deterministic optimal policy*

In the nutshell: if we know the *complete state* and its *transition model*

- We *always* can find a *deterministic optimal policy*
- We might have *multiple deterministic optimal policies*
 - ↳ In that case, we are going to have *also random optimal policies*

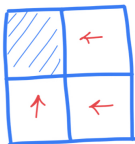
Example: Dummy Grid World



Let's find optimal policy at **state** $s = 3$ in our dummy grid world: first, we write

$$\begin{bmatrix} q_{\star}(3,0) \\ q_{\star}(3,1) \\ q_{\star}(3,2) \\ q_{\star}(3,3) \end{bmatrix} = \begin{bmatrix} \bar{\mathcal{R}}(3,0) + \sum_{\bar{s}} v_{\star}(\bar{s}) p(\bar{s}|3,0) \\ \bar{\mathcal{R}}(3,1) + \sum_{\bar{s}} v_{\star}(\bar{s}) p(\bar{s}|3,1) \\ \bar{\mathcal{R}}(3,2) + \sum_{\bar{s}} v_{\star}(\bar{s}) p(\bar{s}|3,2) \\ \bar{\mathcal{R}}(3,3) + \sum_{\bar{s}} v_{\star}(\bar{s}) p(\bar{s}|3,3) \end{bmatrix} = \begin{bmatrix} -1 & -1 \\ -0.5 & -2 \\ -0.5 & -2 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} -2 \\ -2.5 \\ -2.5 \\ -2 \end{bmatrix}$$

Example: *Dummy Grid World*

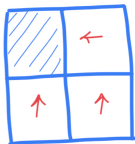


The optimal policy at *state* $s = 3$ is then given by

$$\pi^*(a|3) = \begin{cases} 1 & a = \operatorname{argmax}_a q_\star(3, a) \\ 0 & a \neq \operatorname{argmax}_a q_\star(3, a) \end{cases} = \begin{cases} 1 & a = 0 \\ 0 & a \neq 0 \end{cases}$$

This is obviously *optimal* in this problem!

Example: *Dummy Grid World*

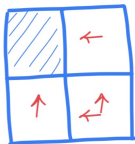


The optimal policy at *state* $s = 3$ is then given by

$$\pi^*(a|3) = \begin{cases} 1 & a = \operatorname{argmax}_a q_\star(3, a) \\ 0 & a \neq \operatorname{argmax}_a q_\star(3, a) \end{cases} = \begin{cases} 1 & a = 3 \\ 0 & a \neq 3 \end{cases}$$

This is obviously *optimal* in this problem!

Example: Dummy Grid World

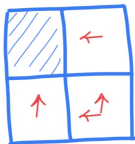


The optimal policy at *state* $s = 3$ is then given by

$$\pi^*(a|3) = \begin{cases} 0.5 & a = 0 \\ 0 & a = 1, 2 \\ 0.5 & a = 3 \end{cases}$$

This is *also optimal* in this problem!

Example: Dummy Grid World



The optimal policy at **state** $s = 3$ is then given by

$$\pi^*(a|3) = \begin{cases} 0.2 & a = 0 \\ 0 & a = 1, 2 \\ 0.8 & a = 3 \end{cases}$$

This is **also optimal** in this problem!